Day 2: mathematical diagnostic

Name: _____ Date: _____

60 minutes total. No calculator, notes, internet or AI on the independent pass. Show your reasoning. If stuck, leave the useful work you have and move on. This is not a speed contest.

3 minutes setup · up to 30 minutes core · up to 20 minutes for two selected follow-ups · 7 minutes for your log. The tutor will supply follow-ups. Do not work through the whole reserve book.

If you finish a block early, check your work or continue to the next block with the tutor. Do not exceed the overall limit.

A1-3

Solve over the reals: $(x^2 - 9)/(x - 3) = 6$. State all restrictions.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A2-3

Find all real roots of $2x^3 - 3x^2 - 8x + 12 = 0$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A3-3

The roots of $x^2 - 5x + 3$ are r and s. Find r^2+s^2 without calculating either root.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A1-2

Solve over the reals: $x(x - 4) = 0$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A2-2

Find the remainder when $P(x)=2x^3 - 3x + 5$ is divided by $x - 2$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A3-2

The roots of $x^2 - 7x + 10$ are r and s. Find r+s and rs without solving for them.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A2-1

State the remainder theorem, factor theorem, and rational-root restriction for an integer-coefficient polynomial with nonzero leading coefficient.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A4-1

For nonzero real x, expand $(x+1/x)^2$. What is x^2+1/x^2 in terms of $t=x+1/x$?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

Session log

Which question was hardest to start, and what did you try?

_____

Was any question familiar, hinted, interrupted, or affected by outside study? Record IDs and details.

_____

Follow-up IDs attempted: _____ / _____